

Name: Key

1. a) In the following matrix-vector multiplication over the real numbers, find the values of a and b .

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ b & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} a \\ 6 \\ 22 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ b & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 14 \\ 6 \\ 2b+18 \end{pmatrix}, \quad \text{so} \quad a = 14$$

$2b+18=22$
 $2b=+4$
 $b=+2$

$$\begin{aligned} a &= 14 \\ b &= 2 \end{aligned}$$

b) Give an example of a real matrix that sends u to v , and state the domain and codomain of the corresponding linear transformation.

$$u = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

There are many possible answers, including:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 5 & 6 & 7 \end{pmatrix}$$

First column needs to be $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. The other four ^{columns} can ~~be any~~ have arbitrary entries.
For all of these, the domain is $\mathbb{R}^5$ and the codomain is $\mathbb{R}^2$.

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2. a) For each of the following matrix operations, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} 4 & 0 & 3 & 3 \\ 1 & 0 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 & 1 & 1 \\ 1 & 2 & 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$$

AB , $A+B$, CD , DC

AB is undefined. ($\#cols(A) \neq \#rows(B)$)

$$A+B = \begin{pmatrix} 4+2 & 0+0 & 3+1 & 3+1 \\ 1+1 & 0+2 & 1+0 & 0+0 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 4 & 4 \\ 2 & 2 & 1 & 0 \end{pmatrix}$$

CD is undefined. ($\#cols(C) \neq \#rows(D)$)

$$DC = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 6 & -2 & -22 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$$

b) Give an example of two matrices whose product equals M ; or explain why this is not possible.

$$M = \begin{pmatrix} 1 & 0 & 3 & 0 \\ 2 & 0 & 6 & 0 \\ 3 & 0 & 9 & 0 \end{pmatrix}$$

There are many correct answers, including:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & 0 \\ 2 & 0 & 6 & 0 \\ 3 & 0 & 9 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 3 & 0 \\ 2 & 0 & 6 & 0 \\ 3 & 0 & 9 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 2 & 0 & 2 & 0 \\ 3 & 0 & 3 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & 0 \\ 2 & 0 & 6 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 3 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$